



Modelling Financial Markets

Seminar 5: Conditional Volatility — GARCH Models



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Acronyms used in this material (1/2)

- ▣ ACF: AutoCorrelation Function
- ▣ AI: Artificial Intelligence
- ▣ AIC: Akaike Information Criterion
- ▣ AR: AutoRegressive (model); in event studies: Abnormal Return
- ▣ ARCH: AutoRegressive Conditional Heteroskedasticity
- ▣ ARMA: AutoRegressive Moving Average (model)
- ▣ BET: Bucharest Exchange Trading (index)
- ▣ BIC: Bayesian Information Criterion
- ▣ BVB: Bursa de Valori Bucureşti (the Bucharest Stock Exchange)
- ▣ DM: Diebold–Mariano (test)
- ▣ EGARCH: Exponential GARCH
- ▣ EWMA: Exponentially Weighted Moving Average
- ▣ GARCH: Generalised AutoRegressive Conditional Heteroskedasticity
- ▣ GARCH-N: GARCH with Normal innovations
- ▣ GED: Generalised Error Distribution
- ▣ GJR: Glosten–Jagannathan–Runkle (asymmetric GARCH)
- ▣ GJR-GARCH: Glosten–Jagannathan–Runkle GARCH
- ▣ GW: Giacomini–White (test)
- ▣ HAC: Heteroskedasticity and Autocorrelation Consistent (standard errors)
- ▣ IGARCH: Integrated GARCH
- ▣ LM: Lagrange Multiplier
- ▣ LR: Likelihood Ratio (test)
- ▣ MA: Moving Average
- ▣ MCS: Model Confidence Set
- ▣ MLE: Maximum Likelihood Estimation
- ▣ MZ: Mincer–Zarnowitz (regression)

Acronyms used in this material (2/2)

- ▣ PIT: Probability Integral Transform
- ▣ QLIKE: Quasi-LIKelihood (loss function)
- ▣ QMLE: Quasi-Maximum Likelihood Estimator
- ▣ QQ: Quantile-Quantile (plot)
- ▣ QSD: Quasi Score-Driven (model)
- ▣ US: United States
- ▣ VIX: CBOE Volatility Index

Today's Question and Route

- Question of the seminar: **can we trust a GARCH volatility estimate, its uncertainty and its forecasts?**
- Step 1
 - ▶ A1: the variance recursion and a 5-day forecast on paper; B1: estimate and diagnose GARCH-t; B2–B3: add asymmetry, S&P 500 then BET; C2: audit an AI answer
- Step 2
 - ▶ A3: uncertainty of long-run volatility and half-life; B7: a bootstrap and a test on the boundary; B5: forecast evaluation
- In class: A1, B1, B2, B3, C2; at home: A3, B7, B5
- The seminar comes before Lecture 5: start with the section “What You Need for Today” (seven slides); A3, B7 and B5 build on the lecture
- What you produce: the A answers, one figure and one table per B exercise, the corrected C2 code, one interpretation sentence per exercise
- Notebook: [open the seminar notebook in Colab](#); each task names its notebook section
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Exercise Map

Exercise	Question	Type	Model
A1	how does the variance update, and what is the 5-day risk?	Solved	–
B1	does GARCH-t fit the S&P 500, and do its residuals pass?	Solved	–
B2	does bad news raise S&P 500 volatility more?	Solved	–
B3	the same question for the BET	Proposed	B2
C2	what is wrong in an AI answer?	Proposed	A1, B1
A3	how precise are long-run volatility and half-life?	Solved	–
B7	is $\alpha + \beta = 1$? a bootstrap and a boundary test	Solved	–
B5	does anything forecast better than GARCH-t?	Solved	–
Homework	one transfer exercise: B4, B6 or B8	Proposed	B1, B5, B7
Extension packs	A2, A4, A5, A6, B5 Extended, C1, C3 (optional)	Solved or Proposed	marked on each task

- **[Solved]**: full solution in the slides and in the notebook, a model to copy; **[Proposed]**: you solve it, the solution is discussed in class
- Part A: derivations on paper; Part B: estimation and inference on real data, each ending with an interpretation question; Part C: open questions and AI critique

Data Used

Series	Price used	Sample	Calendar, filter	Returns
S&P 500	close	2000-01-03 to 2026-09-18	weekdays; 3 unchanged closes dropped	6 714
BET	close	2000-01-05 to 2026-09-18	weekdays; 0 unchanged closes dropped	6 670
Bitcoin	close (BTC/USD)	2014-09-17 to 2026-09-18	7 days a week, no filter	4 384
VIX	close	C3 only	weekdays	–
11 US stocks	adjusted close	C3 only, 1995–2019	aligned with the VIX on common days	–

- Daily prices of the course data, one series per asset (field used: close / adjusted close); returns $r_t = 100 (\ln P_t - \ln P_{t-1})$ in %, each series on its own calendar
- “Unchanged closes”: weekdays whose close equals the previous close, i.e. holidays filled with the last price; no missing values remain after the filter
- Annualisation with the observed frequency A (returns per year): 251.4 (S&P 500), 249.8 (BET), 365.3 (Bitcoin) returns per year

Setup: Run and Verify

```
# notebook section "Data" defines
# load_close(name) and pct_returns(name)
r = pct_returns('sp500')      # 100 * diff(log P)
print(len(r), r.index[0], r.index[-1])
print(r.head(3).round(4))

from arch import arch_model
res = arch_model(r, p=1, q=1,
                 dist='t').fit(dispatch='off')
print(res.params)
```

- Expected: 6 714 returns, 2000-01-04 to 2026-09-18
- First three returns (%): -3.9099, 0.1920, 0.0955
- GARCH-t: $\hat{\alpha} = 0.1226$, $\hat{\beta} = 0.8721$, $\hat{\nu} = 6.26$
- If a count or a value differs, stop and check the file, the field, the dates and the filter
- Packages: numpy, pandas, scipy, statsmodels, matplotlib, arch

What You Need for Today (1/7): From Volatility Clustering to a Variance Equation

- This seminar comes before the lecture: these seven slides give only what the exercises need; depth in [Chapter 5](#) (the lecture after the seminar)
- [Chapter 1](#): daily returns are almost uncorrelated, but $|r_t|$ and r_t^2 are autocorrelated (volatility clustering); so we model the **conditional variance**
 - ▶ $r_t = 100 (\ln P_t - \ln P_{t-1})$: daily log return in %; P_t : closing price of day t
 - ▶ $r_t = \mu + \varepsilon_t$, $\varepsilon_t = \sigma_t z_t$: μ mean return, ε_t shock, z_t i.i.d. with mean 0 and variance 1
 - ▶ $\sigma_t^2 = \text{Var}(r_t \mid \mathcal{F}_{t-1})$: the variance of today's return given $\mathcal{F}_{t-1}$, the information at the close of day $t - 1$
- ARCH(1) [Engle \(1982\)](#): today's variance is a constant plus a reaction to yesterday's squared shock
 - ▶ $\sigma_t^2 = \omega + \alpha \varepsilon_{t-1}^2$
- GARCH(1,1) [Bollerslev \(1986\)](#) adds memory: part of yesterday's variance carries over
 - ▶ $\sigma_t^2 = \omega + \alpha \varepsilon_{t-1}^2 + \beta \sigma_{t-1}^2$
 - ▶ $\omega > 0$: variance floor; $\alpha \geq 0$: reaction to yesterday's news; $\beta \geq 0$: memory of yesterday's variance

What You Need for Today (2/7): Persistence, Forecasts and Units

- **Persistence** $p = \alpha + \beta$: the share of today's deviation from the long-run level that survives to tomorrow
 - ▶ If $p < 1$: long-run (unconditional) variance $\bar{\sigma}^2 = \omega/(1 - p)$, the mean of σ_t^2
- Forecasts revert to $\bar{\sigma}^2$: the gap shrinks by the factor p each day
 - ▶ $\mathbb{E}_t \sigma_{t+h}^2 = \bar{\sigma}^2 + p^{h-1}(\sigma_{t+1}^2 - \bar{\sigma}^2)$
 - ▶ $\mathbb{E}_t = \mathbb{E}[\cdot | \mathcal{F}_t]$; h : horizon in days; σ_{t+1}^2 : tomorrow's variance, already known at the close of t
- **Half-life**: the number of days after which half of a variance shock has gone
 - ▶ $h_{1/2} = \ln 0.5 / \ln p$; e.g. $p = 0.95$ gives $-0.693 / (-0.0513) = 13.5$ days; $h_{1/2} \rightarrow \infty$ as $p \rightarrow 1$
- **Units**: returns in %, so variances in %²
 - ▶ Annual volatility $\sqrt{A \bar{\sigma}^2}$, A : number of returns per year (a square root of a variance, never $A \bar{\sigma}^2$)
 - ▶ $\bar{\sigma}^2 = 1$ and $A = 252$ give $\sqrt{252} = 15.9\%$ a year

What You Need for Today (3/7): Testing for ARCH Effects

- H_0 : no ARCH effects, the conditional variance is constant; both tests ask whether past squared returns predict today's
- **LM test Engle (1982)**: regress $\hat{\varepsilon}_t^2$ on a constant and $\hat{\varepsilon}_{t-1}^2, \dots, \hat{\varepsilon}_{t-q}^2$
 - ▶ $\hat{\varepsilon}_t = r_t - \bar{r}$: demeaned return, $\bar{r}$ the sample mean; q : number of lags (5 in B1)
 - ▶ $LM = nR^2$: n observations, R^2 the coefficient of determination of this regression; χ_q^2 under H_0
- **Ljung-Box on r_t^2 (Chapter 1)**: the joint size of the first m autocorrelations
 - ▶ $Q(m) = n(n+2) \sum_{k=1}^m \hat{\rho}_k^2 / (n-k)$
 - ▶ $\hat{\rho}_k$: sample autocorrelation of r_t^2 at lag k ; m : number of lags (10 in B1); χ_m^2 under H_0
- Reject H_0 when the statistic exceeds the 5% critical value: χ_5^2 : 11.07, χ_{10}^2 : 18.31

What You Need for Today (4/7): Estimating GARCH

- Maximum likelihood: run the recursion for $\sigma_t^2(\theta)$, then choose θ to maximise the log-likelihood
 - ▶ $\ell(\theta) = \sum_t [\ln f(\varepsilon_t/\sigma_t) - \ln \sigma_t]$
 - ▶ θ : all parameters, e.g. $(\mu, \omega, \alpha, \beta, \nu)$; $\varepsilon_t = r_t - \mu$; $\hat{\theta}$: the maximiser
 - ▶ f : density of z_t , Normal or Student-t with ν degrees of freedom scaled to variance 1
 - ▶ A small $\hat{\nu}$ means heavy tails; the fourth moment of z_t is finite only if $\nu > 4$
- Quasi-maximum likelihood: if σ_t^2 is right but f is wrong, the estimates stay consistent
 - ▶ Use the robust (sandwich) covariance $A^{-1}BA^{-1}$ [Bollerslev & Wooldridge \(1992\)](#), not the classical A^{-1}
 - ▶ Here A : minus the mean Hessian (second derivatives of ℓ) per observation; B : variance of the score (first derivatives) per observation
 - ▶ Robust / classical s.e. ratio: 1 if f is right, above 1 with heavier tails than assumed

What You Need for Today (5/7): Checking the Fit

- **Standardised residual** $\hat{z}_t = \hat{\varepsilon}_t / \hat{\sigma}_t$: if the model is right, i.i.d. with mean 0 and variance 1
 - ▶ Ljung–Box on $\hat{z}_t$: mean equation; on $\hat{z}_t^2$ and the LM test: variance equation
 - ▶ QQ plot of $\hat{z}_t$ against the fitted f : the tails
- **Sign-bias test** Engle & Ng (1993): does the sign of yesterday's shock still matter after the model?
 - ▶ $\hat{z}_t^2 = b_0 + b_1 S_{t-1}^- + b_2 S_{t-1}^- \hat{\varepsilon}_{t-1} + b_3 S_{t-1}^+ \hat{\varepsilon}_{t-1} + u_t$
 - ▶ $S_{t-1}^- = 1(\hat{\varepsilon}_{t-1} < 0)$, with $1(\cdot)$ the indicator (1 if true, 0 otherwise); $S_{t-1}^+ = 1 - S_{t-1}^-$; u_t : regression error
 - ▶ b_1 : sign bias; b_2 , b_3 : size bias of negative and positive shocks; significant slopes: asymmetry left
 - ▶ Joint test $nR^2 \sim \chi_3^2$ under H_0 : no asymmetry left (7.81 at 5%)
- Depth: Chapter 5, sections “Estimation and Uncertainty” and “Innovations and Diagnostics”

What You Need for Today (6/7): Asymmetry and News Impact

- Leverage effect ([Chapter 1](#)): bad news raises volatility more than good news; GJR [Glosten, Jagannathan & Runkle \(1993\)](#) gives negative shocks an extra slope γ
 - ▶ $\sigma_t^2 = \omega + (\alpha + \gamma I_{t-1})\varepsilon_{t-1}^2 + \beta\sigma_{t-1}^2$, $I_{t-1} = 1(\varepsilon_{t-1} < 0)$
 - ▶ Slope α for good news, $\alpha + \gamma$ for bad news; $\gamma > 0$: bad news weighs more
 - ▶ With symmetric innovations, persistence $\rho = \alpha + \gamma/2 + \beta$ (a negative shock occurs half of the time)
- News impact curve [Engle & Ng \(1993\)](#): σ_t^2 as a function of ε_{t-1} , with σ_{t-1}^2 held fixed (e.g. at the sample variance)
 - ▶ Symmetric for GARCH, steeper on the left for GJR
 - ▶ Example: $\omega = 0.02$, $\alpha = 0.02$, $\gamma = 0.15$, $\beta = 0.88$, $\sigma_{t-1}^2 = 1$: $\varepsilon_{t-1} = -3$ gives $0.02 + 0.17 \times 9 + 0.88 = 2.43$, $\varepsilon_{t-1} = +3$ gives $0.02 + 0.02 \times 9 + 0.88 = 1.08$

What You Need for Today (7/7): Nested Models and Information Criteria

- Nested models: GARCH is GJR with $\gamma = 0$; the likelihood-ratio statistic compares the two fits
 - ▶ $LR = 2(\ell_1 - \ell_0)$
 - ▶ ℓ_1, ℓ_0 : maximised log-likelihoods of the larger (GJR) and the smaller (GARCH) model; $LR \geq 0$
 - ▶ $LR \sim \chi_1^2$ under H_0 if the density is right and all other parameters are interior; critical values 3.84 (5%), 6.63 (1%)
 - ▶ A parameter on its bound (e.g. $\hat{\alpha} = 0$, since $\alpha \geq 0$) makes the null law non-standard [Andrews \(2001\)](#); B2 shows how to qualify the conclusion
- Information criteria ([Chapter 2](#)): fit penalised by the number of parameters; the lower, the better
 - ▶ $AIC = -2\ell + 2k$, $BIC = -2\ell + k \ln n$; ℓ : maximised log-likelihood, k : number of parameters, n : observations
 - ▶ With one extra parameter, AIC prefers the larger model when $LR > 2$, BIC when $LR > \ln n$
- Depth: [Chapter 5](#), section “Asymmetry and Model Choice”; the summary of all symbols is on the next slide

Notation Used in Every Exercise

- ▣ $r_t = \mu + \varepsilon_t$, $\varepsilon_t = \sigma_t z_t$, z_t i.i.d. with mean 0 and variance 1; $\mathcal{F}_t$: information at the close of day t , $\mathbb{E}_t = \mathbb{E}[\cdot \mid \mathcal{F}_t]$
- ▣ GARCH(1,1) [Bollerslev \(1986\)](#): $\sigma_t^2 = \omega + \alpha \varepsilon_{t-1}^2 + \beta \sigma_{t-1}^2$; persistence $p = \alpha + \beta$; long-run variance $\bar{\sigma}^2 = \omega / (1 - p)$; half-life $h_{1/2} = \ln 0.5 / \ln p$
- ▣ GJR [Glosten, Jagannathan & Runkle \(1993\)](#): add $\gamma I_{t-1} \varepsilon_{t-1}^2$, $I_{t-1} = 1(\varepsilon_{t-1} < 0)$; Student-t innovations with ν degrees of freedom, scaled to variance 1
- ▣ Standardised residual $\hat{z}_t = \hat{\varepsilon}_t / \hat{\sigma}_t$; $LR = 2(\ell_1 - \ell_0)$; $AIC = -2\ell + 2k$, $BIC = -2\ell + k \ln n$ (k parameters)
- ▣ Forecast h_t of the variance of day t , made at the close of $t - 1$; loss $QLIKE = r_t^2 / h_t + \ln h_t$ [Patton \(2011\)](#)
- ▣ One worked line: $\omega = 0.05$, $\alpha = 0.10$, $\beta = 0.85$, $\varepsilon_{t-1} = 2$, $\sigma_{t-1}^2 = 1$: $\sigma_t^2 = 0.05 + 0.40 + 0.85 = 1.30$

A1 [Solved]: The Variance Recursion — Task (1/2)

- **Question:** how does GARCH(1,1) turn four shocks into four daily variances?
- **Inputs:** $\omega = 0.05$, $\alpha = 0.10$, $\beta = 0.85$ (returns in %); start $\sigma_1^2 = \bar{\sigma}^2$; shocks $\varepsilon_1, \dots, \varepsilon_4 = -2.0, 0.5, 3.0, -1.0$; no data file
- **Tasks**
 1. Compute the persistence p and the long-run variance $\bar{\sigma}^2 = \omega / (1 - p)$.
 2. Complete the worksheet below row by row: news term $\alpha \varepsilon_{t-1}^2$, memory term $\beta \sigma_{t-1}^2$ and σ_t^2 for $t = 2, \dots, 5$.
 3. Say which shock moves the variance most, and why the variance of day 5 is known at the close of day 4.
- **Report:** the completed table and one sentence
- **Notebook:** section A1

t	ε_{t-1}	$\alpha \varepsilon_{t-1}^2$	$\beta \sigma_{t-1}^2$	σ_t^2
1	—	—	—	1.0000
2	-2.0	0.4000	0.8500	1.3000
3	0.5	...	...	...
4	3.0	...	...	...
5	-1.0	...	...	...

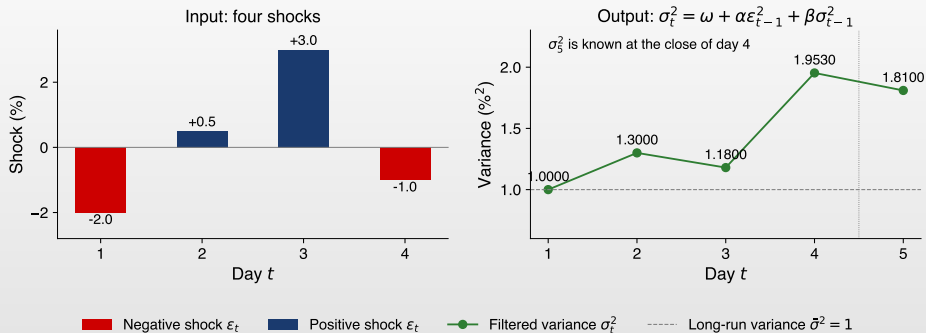
A1 [Solved]: Solution (1/2) — the Filtered Variances

t	ε_{t-1}	$\alpha \varepsilon_{t-1}^2$	$\beta \sigma_{t-1}^2$	σ_t^2
1	—	—	—	1.0000
2	-2.0	0.4000	0.8500	1.3000
3	0.5	0.0250	1.1050	1.1800
4	3.0	0.9000	1.0030	1.9530
5	-1.0	0.1000	1.6600	1.8100

- $p = 0.95$, $\bar{\sigma}^2 = 0.05/0.05 = 1$; each row adds $\omega = 0.05$ to the two terms
- Row 2: $0.05 + 0.10 \times 4 + 0.85 \times 1 = 1.3000$

- The shock of +3% adds 0.90: the variance jumps to 1.9530 on day 4 and decays by β afterwards
- σ_5^2 uses only ε_4 and σ_4^2 , both known at the close of day 4: the one-step forecast is exact within the model
- Interpretation: the sign does not matter, the size does; one large shock raises the variance for many days

A1 [Solved]: the Recursion in a Chart



- Each bar is a shock; each point the variance it produces the next day; σ_5^2 is the one-step forecast made at the close of day 4

A1 [Solved]: Forecasting from Day 4 — Task (2/2)

□ Questions

- ▶ what is the 5-day variance after these shocks?
- ▶ is the square-root-of-time rule right?

□ **Inputs:** the model and $\sigma_5^2 = 1.8100$ from Task 1; forecast origin: the close of day 4

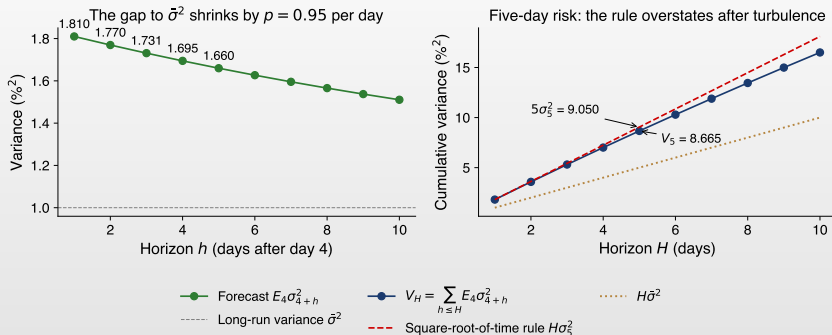
□ Tasks

1. Take $\mathbb{E}_4$ of the recursion and use $\mathbb{E}_4 \varepsilon_t^2 = \mathbb{E}_4 \sigma_t^2$ for $t \geq 5$ to derive $\mathbb{E}_4 \sigma_{4+h}^2 = \bar{\sigma}^2 + p^{h-1}(\sigma_5^2 - \bar{\sigma}^2)$.
 2. Compute the five forecasts for $h = 1, \dots, 5$.
 3. Explain why the 5-day return variance is the sum $V_5 = \sum_{h=1}^5 \mathbb{E}_4 \sigma_{4+h}^2$ (hint: the ε_t are uncorrelated, a martingale difference), and compute V_5 and $\sqrt{V_5}$.
 4. Compare V_5 with $5\sigma_5^2$ and derive the sign of $V_H - H\sigma_5^2$ as a function of $\sigma_5^2 - \bar{\sigma}^2$.
 5. Compute the half-life $\ln 0.5 / \ln p$.
- **Report:** the forecast formula, five forecasts, V_5 and $\sqrt{V_5}$, the half-life; interpretation: after a +3% shock, does the square-root-of-time rule overstate or understate the 5-day risk?
- **Notebook:** section A1

A1 [Solved]: Solution (2/2) — Forecasts and Five-Day Risk

- Derivation: $\mathbb{E}_4 \sigma_{4+h}^2 = \omega + p \mathbb{E}_4 \sigma_{4+h-1}^2$; subtract $\bar{\sigma}^2 = \omega + p \bar{\sigma}^2$: the gap shrinks by p per day, starting from the known σ_5^2
- Gap 0.8101; forecasts 1.8100, 1.7695, 1.7311, 1.6945, 1.6598; $V_5 = 5 + 0.8101 \times (1 - 0.95^5)/0.05 = 8.665$, $\sqrt{V_5} = 2.944\%$; half-life $-0.6931/(-0.0513) = 13.5$ days
- $V_H - H\sigma_5^2 = (\bar{\sigma}^2 - \sigma_5^2) \sum_{j < H} (1 - p^j)$: here $5\sigma_5^2 = 9.050 > V_5$. Interpretation: after turbulence the rule overstates the 5-day risk, in calm times it understates it, because the variance reverts to $\bar{\sigma}^2$

A1 [Solved]: Forecasts and Five-Day Risk in a Chart



Left: the forecasts converge to $\bar{\sigma}^2 = 1$; right: V_H stays below $H \sigma_5^2$ because the variance reverts

B1 [Solved]: GARCH-t on the S&P 500 — Task (1/2)

Questions

- ▶ is there clustering to model?
- ▶ what does a GARCH(1,1)-t estimate say about it?

Inputs: S&P 500, 6 714 daily returns (Data Used); mean $r_t = \mu + \varepsilon_t$, Student-t innovations

Tasks

1. Test H_0 : no ARCH effects with Ljung–Box $Q(10)$ on r_t^2 and Engle's LM test with 5 lags on $r_t - \bar{r}$, at 5%.
2. Estimate GARCH(1,1)-t by maximum likelihood and check that the optimiser converged.
3. Report every parameter with its robust [Bollerslev & Wooldridge \(1992\)](#) and classical standard error, and plot r_t with the bands $\pm 2\hat{\sigma}_t$.

Report: the two test decisions, the estimates table, the chart

Notebook: section B1

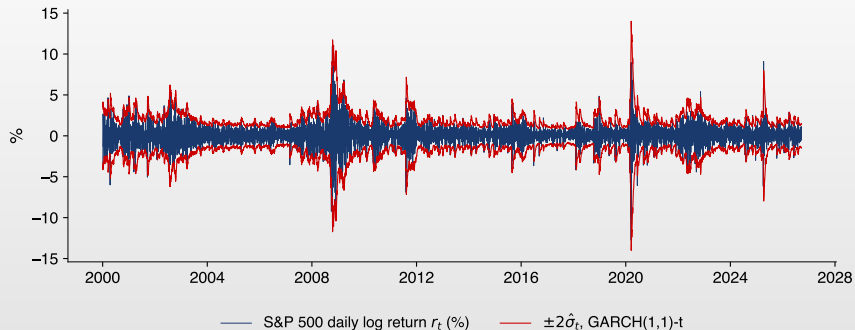
```
from arch import arch_model
am = arch_model(r, mean='Constant',
                vol='GARCH', p=1, q=1,
                dist='t')
res = am.fit(dispatch='off')
print(res.convergence_flag) # 0 = ok
rob = res.std_err           # robust
cl = am.fit(dispatch='off',
            cov_type='classic').std_err
sig = res.conditional_volatility
```

B1 [Solved]: Solution (1/2) — Tests, Estimates, Fit

Parameter	Estimate	Robust s.e.	Classical s.e.
μ	0.0788	0.0089	0.0090
ω	0.0157	0.0032	0.0029
α	0.1226	0.0107	0.0101
β	0.8721	0.0103	0.0096
ν	6.26	0.48	0.48

- (1) $Q(10) = 6133$ and $LM(5) = 1684$ against 18.3 and 11.1: no-ARCH is rejected ($p < 0.001$)
- (2) Converged after 14 iterations, $\ell = -9062.5$; persistence 0.9947
- (3) Robust/classical ratios 1.06 (α), 1.08 (β): the Student-t density is itself misspecified (skewness -0.58); the sandwich covers that

B1 [Solved]: the Fitted Volatility



□ Interpretation: $\hat{\beta}$ carries yesterday's variance, $\hat{\alpha}$ adds the news, $\hat{\nu}$ signals heavy tails; 5.1% of the days fall outside $\pm 2\hat{\sigma}_t$

B1 [Solved]: Diagnosing the Residuals — Task (2/2)

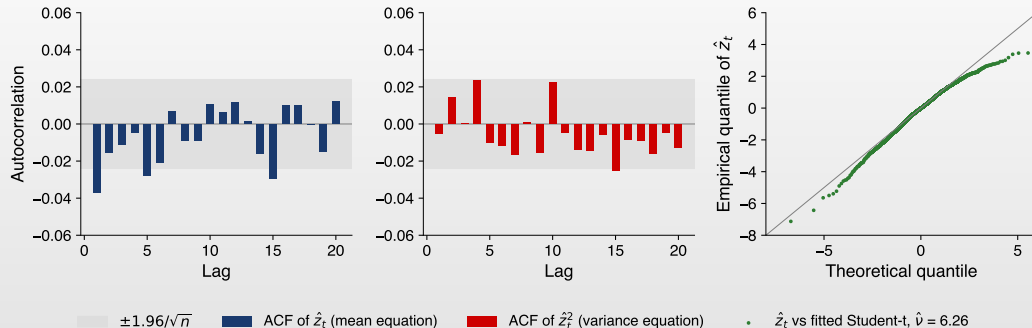
- **Question:** does the fitted GARCH-t leave any structure in $\hat{z}_t = (r_t - \hat{\mu})/\hat{\sigma}_t$?
- **Inputs:** the fit of Task 1; each test has H_0 : the model captures that feature
- **Tasks**
 1. Test the mean equation with Ljung–Box $Q(10)$ on $\hat{z}_t$ (H_0 : no autocorrelation up to lag 10).
 2. Test the variance equation with $Q(10)$ on $\hat{z}_t^2$ and the LM(5) test (H_0 : no clustering left).
 3. Run the sign-bias regression [Engle & Ng \(1993\)](#) $\hat{z}_t^2 = b_0 + b_1 S_{t-1}^- + b_2 S_{t-1}^- \hat{\varepsilon}_{t-1} + b_3 S_{t-1}^+ \hat{\varepsilon}_{t-1}$, with $S_{t-1}^- = 1(\hat{\varepsilon}_{t-1} < 0)$ and $S_{t-1}^+ = 1 - S_{t-1}^-$; report the three t -statistics and the joint $nR^2 \sim \chi_3^2$ (H_0 : no asymmetry left).
 4. Plot the ACFs of $\hat{z}_t$ and $\hat{z}_t^2$ and a QQ plot of $\hat{z}_t$ against the fitted standardised Student-t.
- **Report:** a table with statistic, p -value and decision for each test, and the three panels; interpretation:
 - ▶ which part of the model fails?
 - ▶ what would you change first?
- **Notebook:** section B1

B1 [Solved]: Solution (2/2) — Diagnostics

Part	Test	p	Decision at 5%
Mean	$Q(10), \hat{z}: 21.9$	0.016	reject
Variance	$Q(10), \hat{z}^2: 13.6$	0.192	not rejected
	$LM(5): 5.8$	0.326	not rejected
Asymmetry	joint $\chi^2_3: 40.7$	< 0.001	reject

- Sign bias $t = 3.56$, negative size $t = 1.13$, positive size $t = -2.61$
- QQ: 16 residuals below -4 against 8.5 expected; the left tail is longer than the fitted t

B1 [Solved]: Residual ACFs and QQ Plot



- All p -values use the naive χ^2 laws, approximate with estimated parameters [Li & Mak \(1994\)](#). Interpretation: clustering is captured, asymmetry is not: add a sign term (B2)

B2 [Solved]: GJR versus GARCH on the S&P 500 — Task

▣ Questions

- ▶ does bad news raise S&P 500 volatility more than good news?
- ▶ by how much?

- ▣ **Inputs:** B1 data; `arch_model(r, p=1, o=1, q=1, dist='t')` estimates GJR-GARCH(1,1)-t, which nests GARCH-t at $\gamma = 0$; news impact at $\sigma_{t-1}^2 = \text{sample variance } 1.47$

▣ Tasks

1. Estimate GJR-GARCH(1,1)-t and report $\hat{\alpha}$, $\hat{\gamma}$ with its robust t , and $\hat{\beta}$.
2. Build the comparison table: number of parameters k , ℓ , AIC, BIC for GARCH-t and GJR-t.
3. Test $H_0: \gamma = 0$ with $LR = 2(\ell_{GJR} - \ell_{GARCH})$ against χ_1^2 and state the assumptions behind this law.
4. Compute the next-day variance after shocks of -3% and $+3\%$ in both models and plot the two news impact curves.

- ▣ **Report:** the estimates, the comparison table, the LR decision and the news-impact chart; interpretation:

- ▶ why is $\hat{\alpha} = 0$?
- ▶ what does it do to the χ_1^2 law?

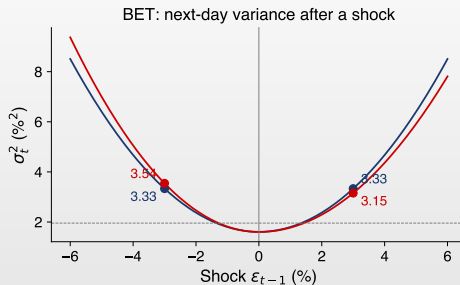
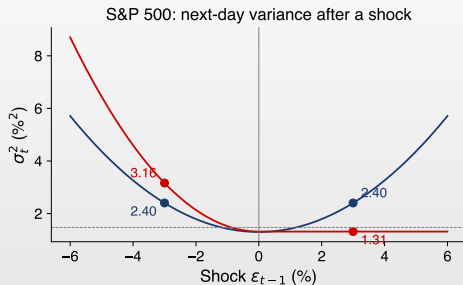
- ▣ Notebook: section B2

B2 [Solved]: Solution (1/2) — Nested Comparison

Model	k	ℓ	AIC	BIC
GARCH(1,1)-t	5	-9062.473	18134.9	18169.0
GJR-GARCH(1,1)-t	6	-8944.647	17901.3	17942.2

- (1) $\hat{\alpha} = 0.000$, $\hat{\gamma} = 0.205$ (robust $t = 10.03$), $\hat{\beta} = 0.881$, $\hat{\omega} = 0.0187$, $\hat{\nu} = 6.96$: good news slope $\hat{\alpha}$, bad news $\hat{\alpha} + \hat{\gamma}$
- (2) $\Delta\text{AIC} = -233.7$, $\Delta\text{BIC} = -226.8$ (GJR minus GARCH): both prefer GJR
- (3) $LR = 2(-8944.647 - (-9062.473)) = 235.65$, far above 3.84
- Assumptions of the χ^2_1 law: a correct Student-t density and all other parameters interior
 - ▶ Both are doubtful here: B1 found skewness, and $\hat{\alpha} = 0$ sits on the bound $\alpha \geq 0$, so the law is non-standard [Andrews \(2001\)](#)
 - ▶ Qualified conclusion: an LR of 235.65 is some 60 times the 5% critical value; a calibrated check would simulate the LR under the fitted GARCH-t (as B7 does for IGARCH)

B2 [Solved]: Solution (2/2) — News Impact



- (4) S&P 500, GJR: -3% gives 3.16, $+3\%$ gives 1.31 (2.40 times less); GARCH: 2.40 for both signs
- With $\hat{\alpha} = 0$ a $+3\%$ shock even lowers the variance, from 1.47 to 1.31. Interpretation: in the S&P 500, only bad news raises tomorrow's variance; the BET panel is B3

B3 [Proposed]: GJR versus GARCH on the BET — Task

- **Question:** is the asymmetry of the Bucharest market as strong as that of the S&P 500?
- **Inputs:** BET, 6 670 daily returns (Data Used); the same two models and code as B2; $\ln n = 8.81$
- **Tasks**
 1. Estimate both models and fill the comparison table of B2 (k , ℓ , AIC, BIC) for the BET.
 2. Compute LR and place it against the four thresholds: 2 (AIC), 3.84 and 6.63 (χ^2_1 at 5% and 1%) and $\ln n$ (BIC); state each decision.
 3. Compute the BET news impact at $\pm 3\%$ from the sample variance, as in B2.
- **Report:** the table, the four decisions, the two news-impact values; interpretation:
 - ▶ why can LR and BIC disagree?
 - ▶ does a smaller $\hat{\gamma}$ prove that the BET is less asymmetric than the S&P 500?
- **Model:** B2
- **Notebook:** section B3

C2 [Proposed]: Critique an AI Answer — Task

- **Prompt:** “px holds daily S&P 500 closes. Fit a GARCH(1,1) with Student-*t* innovations using arch and give me the annualised long-run volatility and the half-life of a shock.”
- **Input:** px = load_close('sp500'), 6 715 closes, 2000-01-03 to 2026-09-18
- **Answer's output:** ann_vol = 66.46, half_life = 0.74 days, and a DataScaleWarning
- **Tasks**
 1. Run the code and reproduce the output.
 2. Find the formula errors, the unit errors and the numerical risks, each with its line number.
 3. For each, give the evidence that exposes it (a warning, a number from A1 or B1) and the corrected line.
 4. Run the corrected code and log each error as in AI_ERRORS.md.
- **Model:** A1, B1; annualisation: the primer and the slide “Data Used”

```
1 import numpy as np
2 from arch import arch_model
3 r = np.log(px).diff().dropna()
4 am = arch_model(r, vol='GARCH', p=1,
5                 q=1, dist='t')
6 res = am.fit(disp='off')
7 om = res.params['omega']
8 a = res.params['alpha[1]']
9 b = res.params['beta[1]']
10 lr_var = om / (1 - a - b)
11 ann_vol = lr_var * 252
12 half_life = np.log(0.5) / np.log(b)
```

Tools for A3, B7 and B5 (1/2): Delta Method and Parametric Bootstrap

- **Delta method:** the standard error of a smooth function $g(\hat{\theta})$ of the estimates, by a first-order (linear) approximation
 - ▶ $\widehat{\text{s.e.}} g(\hat{\theta}) \approx \sqrt{\nabla g^\top \hat{V} \nabla g}$
 - ▶ ∇g : gradient of g at $\hat{\theta}$ (column of partial derivatives), $\nabla g^\top$ its transpose; $\hat{V}$: estimated (robust) covariance matrix of $\hat{\theta}$
 - ▶ Reliable only if g is nearly linear over the uncertainty range of $\hat{\theta}$ (A3 shows where it fails)
- **Parametric bootstrap:** simulate B paths from a fitted model and re-estimate on each; starred quantities ($\hat{p}^*$, LR^*) come from the simulated paths
 - ▶ Percentile interval: the 2.5% and 97.5% percentiles of the B values $\hat{p}^*$
 - ▶ Bootstrap test: simulate under H_0 ; $p = (1 + \#\{LR^* \geq LR\})/(B + 1)$, $\#\{\cdot\}$: number of draws that satisfy the condition
 - ▶ Boundary law $\frac{1}{2}\chi_0^2 + \frac{1}{2}\chi_1^2$: $LR = 0$ with probability $\frac{1}{2}$, otherwise χ_1^2 ; 5% critical value 2.71

Tools for A3, B7 and B5 (2/2): Comparing Variance Forecasts

- h_t : variance forecast for day t , made at the close of $t - 1$; the true variance is never observed, so r_t^2 serves as a noisy proxy
- **QLIKE loss** Patton (2011): $L_t = r_t^2/h_t + \ln h_t$
 - ▶ Its expected value is lowest at the true variance; lower mean QLIKE is better; under-prediction is punished more
- **Diebold–Mariano** Diebold & Mariano (1995): $d_t = L_t^m - L_t^{GARCH-t}$, $DM = \bar{d}/\widehat{\text{s.e.}}(\bar{d})$
 - ▶ L_t^m : loss of model m ; $\bar{d}$: mean of d_t ; HAC s.e. (robust to heteroskedasticity and autocorrelation, 5 lags)
 - ▶ $\approx N(0, 1)$ under equal accuracy; $DM < -1.96$: model m better than GARCH-t, $DM > 1.96$: worse
- **Mincer–Zarnowitz** Mincer & Zarnowitz (1969): $r_t^2 = a + b h_t + u_t$; unbiased forecasts: $a = 0$, $b = 1$
 - ▶ Joint Wald test of $a = 0$, $b = 1$ (χ_2^2); R^2 : share of the variation of r_t^2 explained, low because the proxy is noisy
- Benchmarks: EWMA $\sigma_t^2 = \lambda \sigma_{t-1}^2 + (1 - \lambda) r_{t-1}^2$ (λ : decay factor); Hist-20: mean of the last 20 r_t^2

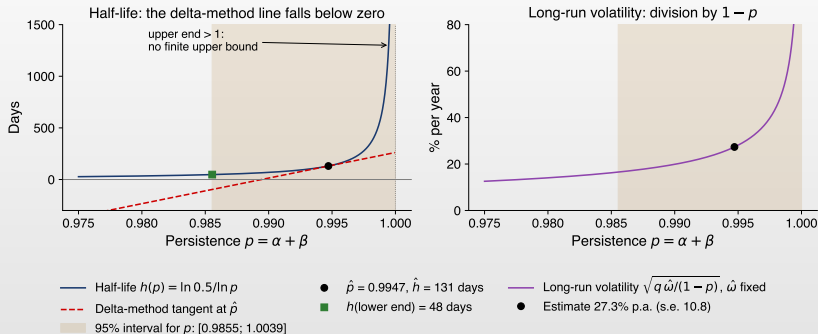
A3 [Solved]: Long-Run Volatility and Half-Life — Task

- ▣ **Question:** how precisely does the B1 fit pin down the long-run volatility and the half-life?
- ▣ **Inputs:** B1: $\hat{\omega} = 0.015714$, $\hat{\alpha} = 0.122573$, $\hat{\beta} = 0.872139$, $A = 251.4$ returns a year; robust covariance of $(\hat{\omega}, \hat{\alpha}, \hat{\beta})$, in this order: $\hat{V} = \begin{pmatrix} 1.001 \times 10^{-5} & 1.421 \times 10^{-5} & -2.242 \times 10^{-5} \\ . & 1.153 \times 10^{-4} & -1.002 \times 10^{-4} \\ . & . & 1.069 \times 10^{-4} \end{pmatrix}$ (res.param_cov)
- ▣ **Tasks**
 1. Derive $\bar{\sigma}^2 = \omega / (1 - \alpha - \beta)$ from the recursion and compute the long-run annual volatility $v = \sqrt{A \bar{\sigma}^2}$ and the half-life.
 2. Compute $\nabla \bar{\sigma}^2$, the quadratic form $\nabla' \hat{V} \nabla$ and, with $\partial v / \partial \bar{\sigma}^2 = A / (2v)$, the standard error of v .
 3. Compute $\widehat{\text{s.e.}}(\hat{p})$ from $\hat{V}$ and the delta-method standard error of $h(p) = \ln 0.5 / \ln p$, defined for $0 < p < 1$.
 4. Map the 95% interval for p into an interval for h and say what happens when it crosses 1.
- ▣ **Report:** v with its s.e., h with its s.e. and the transformed interval; interpretation: why does the linearisation fail for h ?
- ▣ **Notebook:** section A3

A3 [Solved]: Solution — Gradients and the Half-Life Interval

- (1) $\hat{p} = 0.994712$, $\bar{\sigma}^2 = 2.972$, $v = 1.724 \times \sqrt{251.4} = 27.3\%$ (sample: 19.2%); $h = -0.6931/(-0.005302) = 130.7$ days
- (2) $\nabla \bar{\sigma}^2 = (1/(1-p), \omega/(1-p)^2, \omega/(1-p)^2) = (189.1, 562.0, 562.0)$; $\widehat{\text{s.e.}}(\bar{\sigma}^2) = 2.35$; $\widehat{\text{s.e.}}(v) = 10.8$ points around 27.3%
- (3) $\widehat{\text{s.e.}}(\hat{p}) = \sqrt{V_{\alpha\alpha} + 2V_{\alpha\beta} + V_{\beta\beta}} = 0.0047$ (V_{ij} : entries of $\hat{V}$); $\partial h / \partial p = 24\,789$, so $\widehat{\text{s.e.}}(\hat{h}) = 116$ days
- (4) $[0.9855; 1.0039]$ crosses 1: $h \in [48; \infty)$, no finite upper bound within the stationary model. Interpretation: $h(p)$ is so curved near 1 that its tangent is meaningless; the question $p = 1$ needs a test (B7)

A3 [Solved]: the Half-Life Curve



- The tangent (delta method) crosses zero inside the interval for p ; the exact map $h(p)$ has no finite upper end

B7 [Solved]: Is $\alpha + \beta = 1$? Bootstrap and Boundary Test — Task

▣ Questions

- ▶ how variable is $\hat{p} = \hat{\alpha} + \hat{\beta}$?
- ▶ can the S&P 500 variance have a unit root?

- ▣ **Inputs:** B1 fit, $\hat{p} = 0.9947$; code `param_bootstrap`, `igarch_test` (notebook); seed 42 (43 under H_0); each path: 6 714 days after 500 burn-in days, started at the sample variance; each fit imposes $\alpha + \beta \leq 1$ and keeps the best of two starting points

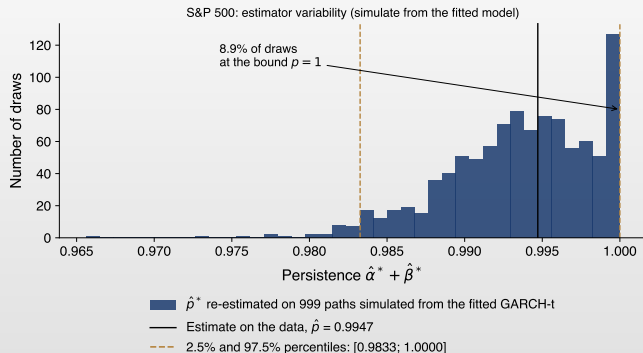
▣ Tasks

1. Algorithm 1 (estimator variability): simulate $B = 999$ paths from the **fitted** GARCH-t, re-estimate p on each, and report the 2.5% and 97.5% percentiles and the share of draws at $p = 1$.
2. Algorithm 2 (test): fit IGARCH-t ($\alpha + \beta = 1$), compute $LR = 2(\ell_{GARCH} - \ell_{IGARCH})$, simulate B paths **under** H_0 , re-estimate both models on each path, and compute $p = (1 + \#\{LR^* \geq LR\})/(B + 1)$.
3. Compare this p -value with the χ_1^2 and $\frac{1}{2}\chi_0^2 + \frac{1}{2}\chi_1^2$ values and decide at 5% and 10%.

- ▣ **Report:** two histograms, the interval, three p -values and the decision; interpretation: why is “the percentile interval reaches 1” not a test of $p = 1$ [Andrews \(2000\)](#)?

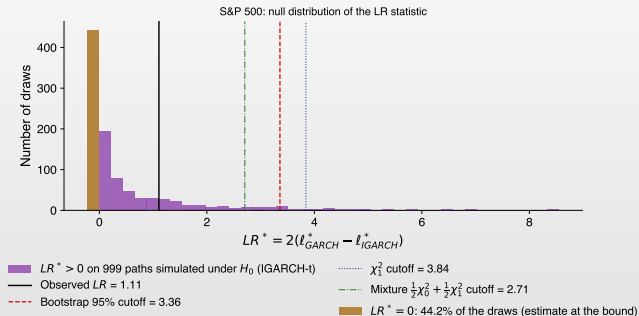
- ▣ Notebook: section B7

B7 [Solved]: Solution (1/2) — Estimator Variability



- 95% percentile interval [0.9833; 1.0000]; bootstrap s.d. 0.0047 against the robust s.e. 0.0047: close; 8.9% of the draws sit exactly at the bound $\rho = 1$
- Half-life from 41 days to ∞ (delta method: 116 days, A3). Interpretation: near a bound the percentile interval describes spread, it is not automatically a valid confidence interval [Andrews \(2000\)](#)

B7 [Solved]: Solution (2/2) — the Test under the Null



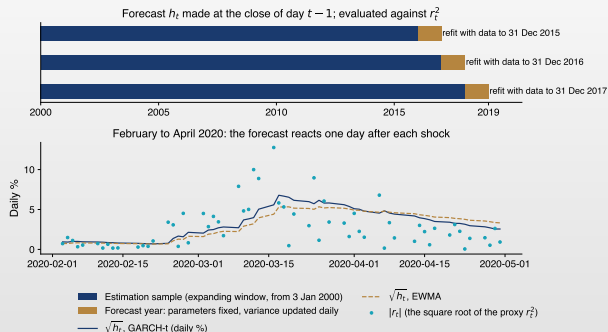
- IGARCH-t: $\hat{\alpha}_0 = 0.127$, $\hat{\nu}_0 = 6.03$;
 $LR = 1.11$; $LR^* = 0$ exactly for 44.2% of the draws (close to the half the mixture predicts; the lecture quotes the share of re-estimates with $\hat{\alpha} + \hat{\beta} \geq 0.999$, 52%); 174 of 999 draws exceed LR
- $p = 0.17$ (bootstrap), 0.15 (mixture), 0.29 (naive χ_1^2): H_0 is not rejected at 5% or 10%. Interpretation: 27 years cannot tell a half-life of about 130 days from an infinite one

B5 [Solved]: Forecast Comparison on the S&P 500 — Task (1/2)

- **Question:** does any model forecast tomorrow's S&P 500 variance better than GARCH-t?
- **Inputs:** 2 692 forecast days, 2016-01-04 to 2026-09-18; GARCH-N, GARCH-t, GJR-t, EGARCH-t refit each January (expanding window); EWMA ($\lambda = 0.94$) [J.P. Morgan/Reuters \(1996\)](#); Hist-20 (mean of the last 20 r_t^2); proxy r_t^2
- **Tasks**
 1. On the timeline chart, check that h_t uses only data up to the close of $t - 1$ and that parameters are fixed within a year.
 2. Compute $QLIKE = r_t^2/h_t + \ln h_t$ step by step for GARCH-t and EWMA on 2020-03-13, 2020-03-16 and 2020-03-17 (table below).
 3. Compute the mean QLIKE of all six models and the DM t of $d_t = L_t^m - L_t^{GARCH-t}$ (HAC, 5 lags) [Diebold & Mariano \(1995\)](#); plot the cumulative $\sum d_t$.
- **Report:** the three QLIKE rows, a table of mean QLIKE and DM t , the loss chart; interpretation: which differences can be read as significant?
- **Notebook:** section B5

Day	r_t	r_t^2	h_t GARCH-t	h_t EWMA	QLIKE
2020-03-13	8.88	78.87	25.38	15.66	...
2020-03-16	-12.77	162.95	31.18	19.45	...
2020-03-17	5.82	33.90	46.06	28.06	...

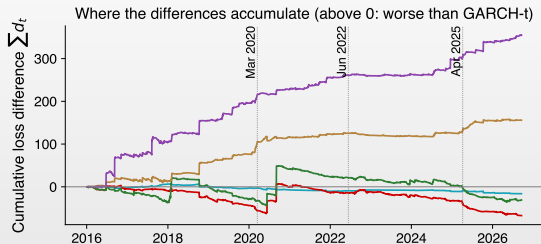
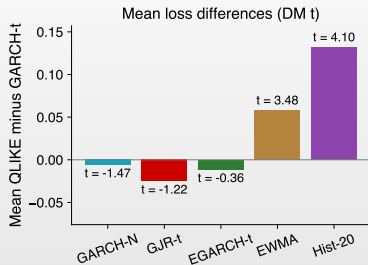
B5 [Solved]: Solution (1/3) — Timing and QLIKE Step by Step



Day	L GARCH-t	L EWMA
2020-03-13	6.342	7.789
2020-03-16	8.666	11.346
2020-03-17	4.566	4.543

- e.g. 2020-03-16:
 $162.95/31.18 + \ln 31.18 = 8.666$; EWMA forecast lower, so a larger loss: under-prediction is punished
- First target 2016-01-04, origin 2015-12-31: refit on data to 31 December, no look-ahead

B5 [Solved]: Solution (2/3) — Loss Differences



— GARCH-N minus GARCH-t — GJR-t minus GARCH-t — EGARCH-t minus GARCH-t — EWMA minus GARCH-t — Hist-20 minus GARCH-t

- Mean QLIKE: GJR-t 0.7319, GARCH-t 0.7570; EWMA (DM $t = 3.48$) and Hist-20 ($t = 4.10$) are worse, their gap grows in March 2020; GARCH-family t are descriptive (nested, expanding window: B5 Extended)

B5 [Solved]: Calibration — Task (2/2)

▣ Questions

- ▶ are the GARCH-t forecasts unbiased in levels?
- ▶ which days drive the answer?

▣ Inputs: the 2 692 pairs (h_t, r_t^2) of Task 1

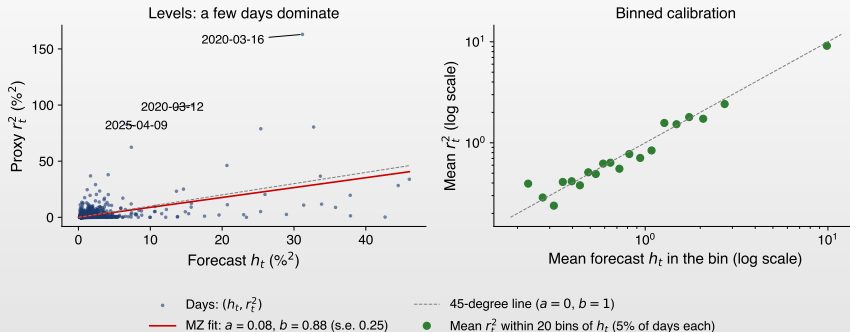
▣ Tasks

1. Estimate the Mincer–Zarnowitz regression $r_t^2 = a + b h_t + u_t$ [Mincer & Zarnowitz \(1969\)](#) with HAC errors (5 lags) and test $a = 0$, $b = 1$ jointly.
2. Plot r_t^2 against h_t with the fitted line and the 45-degree line, and a chart of the means over 20 bins of h_t .
3. Re-estimate without the three largest r_t^2 and compare b .

▣ Report: a , b with s.e., R^2 , the Wald p , the chart, b without the three days; interpretation: is a low R^2 a sign of a bad forecast?

▣ Notebook: section B5

B5 [Solved]: Solution (3/3) — Calibration



- $a = 0.08$ (s.e. 0.24), $b = 0.88$ (s.e. 0.25), $R^2 = 0.238$, Wald $p = 0.804$: unbiasedness is not rejected; binned means lie close to the 45-degree line
- Without 2020-03-16, 2020-03-12, 2025-04-09: $b = 0.64$ (s.e. 0.09), Wald $p < 0.001$. Interpretation: three crisis days carry the level regression; low R^2 reflects the noisy proxy r_t^2 , not a bad model [Patton \(2011\)](#)

Transfer Exercise: Choose B4, B6 or B8

- Choose one exercise from the extension packs and solve it with the code of its model
 - ▶ B4: innovation laws for Bitcoin (model: B1, B2); B6: forecast comparison on the BET (model: B5); B8: bootstrap and IGARCH test on the BET (model: B7)
- What to produce
 - ▶ The table and the chart the task asks for, the answer to the interpretation question and one limitation
- Self-check
 - ▶ Reproducibility: the notebook runs top to bottom and reproduces every reported number
 - ▶ Correct inference: the right null law (boundary, nested models), uncertainty reported with every estimate
 - ▶ Interpretation: a claim supported by the numbers, with its limitation
- AI use is allowed; note the errors you actually catch, in the format of `AI_ERRORS.md`

Exit Check

Exit check (answer in two lines)

- ▣ One justified conclusion: which model would you use for the S&P 500 and why?
- ▣ One unresolved limitation: what can the data not decide (e.g. $p = 1$)?

Extension Packs: What Each One Needs

Exercise	Question	Type	Model
A2	GJR persistence when the innovations are skewed	Proposed	A1
A4	is the fourth moment finite?	Proposed	A1, A3
A5	robust versus classical variance of the Gaussian QMLE	Proposed	A3
A6	autocorrelation of squared shocks	Proposed	A1
B4	innovation laws for Bitcoin	Proposed	B1, B2
B5 Extended	Giacomini–White and the Model Confidence Set	Solved	–
B6	forecast comparison on the BET	Proposed	B5
B8	bootstrap and IGARCH test on the BET	Proposed	B7
C1	does US news drive BET volatility? (project)	Proposed	B1, B2
C3	quasi score-driven GARCH on US stocks (project)	Proposed	B1, B2

- Each pack follows the same format: question, inputs, numbered tasks, report; solutions of proposed exercises are discussed in class

A2 [Proposed]: GJR Persistence with Skewed Innovations — Task

- ▣ **Question:** what is the persistence of GJR-GARCH when z_t is skewed?
- ▣ **Inputs:** S&P 500, GJR-GARCH(1,1) with Hansen skewed-t innovations [Hansen \(1994\)](#): $\hat{\alpha} = 0.00000$, $\hat{\gamma} = 0.21116$, $\hat{\beta} = 0.87965$, $\hat{\nu} = 7.70$, $\hat{\lambda} = -0.150$; the standardised density f of z_t is `SkewStudent().loglikelihood` in `arch`; λ : skewness parameter
- ▣ **Tasks**
 1. Take expectations in the recursion and show that the persistence is $\alpha + \gamma m_- + \beta$, with $m_- = \mathbb{E}[z_t^2 \mathbf{1}(z_t < 0)]$.
 2. Show that $m_- = 1/2$ for a symmetric z_t and explain why m_- differs from $P(z_t < 0)$ in general.
 3. Integrate numerically `quad(f, -inf, 0)` and `quad(lambda z: z**2*f(z), -inf, 0)`; check $\int f = 1$ and $\int z^2 f = 1$ over the real line.
 4. Compute the persistence and the half-life with m_- and with $1/2$.
- ▣ **Report:** m_- , $P(z < 0)$, both persistences and half-lives; interpretation: what goes wrong if $\alpha + \gamma/2 + \beta$ is used with skewed innovations?
- ▣ **Model:** A1
- ▣ **Notebook:** section A2

A4 [Proposed]: Is the Fourth Moment Finite? — Task

▣ Questions

- ▶ do the fitted S&P 500 models have a finite kurtosis?
- ▶ can the sample kurtosis validate them?

- ▣ **Inputs:** GARCH-N: $\hat{\alpha} = 0.120604$, $\hat{\beta} = 0.859878$; GARCH-t: $\hat{\alpha} = 0.1226$, $\hat{\beta} = 0.8721$, $\hat{\nu} = 6.256$; sample kurtosis of the returns 13.7 (total, not excess)

▣ Tasks

1. With $\kappa_z = \mathbb{E}z_t^4$ (the kurtosis of z_t), show that $\mathbb{E}\varepsilon_t^4 = \kappa_z \mathbb{E}\sigma_t^4$ and $\mathbb{E}\sigma_t^4 < \infty$ iff $\mathbb{E}(\alpha z_t^2 + \beta)^2 = (\alpha + \beta)^2 + (\kappa_z - 1)\alpha^2 < 1$.
2. For GARCH-N ($\kappa_z = 3$) compute the condition and the total kurtosis $K = 3[1 - (\alpha + \beta)^2]/[1 - (\alpha + \beta)^2 - 2\alpha^2]$, with the denominator shown; repeat with estimates rounded to 3 decimals.
3. For GARCH-t compute $\kappa_z = 3(\nu - 2)/(\nu - 4)$ and the condition.

- ▣ **Report:** a worksheet row per model: κ_z , the condition, finite or not, K ; interpretation: is agreement with the sample kurtosis a test of the model?

- ▣ Model: A1, A3

- ▣ Notebook: section A4

A5 [Proposed]: Robust versus Classical Variance, ARCH(1) — Task

- **Question:** by how much does the inverse Hessian misstate the variance of the Gaussian QMLE when z_t is not Normal?
- **Inputs:** ARCH(1), $\sigma_t^2 = \omega + \alpha \varepsilon_{t-1}^2$, ω known; $\ell_t = -\frac{1}{2}(\ln \sigma_t^2 + \varepsilon_t^2 / \sigma_t^2)$; $x_t = \varepsilon_{t-1}^2 / \sigma_t^2$ is known at $t-1$ and independent of z_t ; identities $\partial \sigma_t^2 / \partial \alpha = \varepsilon_{t-1}^2$, $\varepsilon_t^2 / \sigma_t^2 = z_t^2$
- **Tasks**
 1. Show that the score is $s_t = \frac{1}{2}(z_t^2 - 1)x_t$ and that $B = \mathbb{E}s_t^2 = \frac{1}{4}(\kappa_z - 1)\mathbb{E}x_t^2$.
 2. Show that $\partial^2 \ell_t / \partial \alpha^2 = \frac{1}{2}x_t^2(1 - 2z_t^2)$ and $A = \frac{1}{2}\mathbb{E}x_t^2$.
 3. Show that the asymptotic variance of $\sqrt{n}(\hat{\alpha} - \alpha)$ is $A^{-1}BA^{-1} = \frac{\kappa_z - 1}{2}A^{-1}$; the finite-sample variance divides it by n .
 4. Compute the s.e. ratio $\sqrt{(\kappa_z - 1)/2}$ for Student-t with $\nu = 6.26$ and for $\hat{\kappa}_z = 4.73$.
- **Report:** s_t , A , B , the ratio, two numbers; interpretation: what happens to the $\sqrt{n}$ theory when $\kappa_z = \infty$?
- **Model:** A3
- **Notebook:** section A5

A6 [Proposed]: Autocorrelation of Squared Shocks — Task

▣ Questions

- ▶ does the ACF of r_t^2 implied by GARCH-N match the data?
- ▶ how precise is the sample ACF?

- ▣ **Inputs:** ARMA(1,1) form $\varepsilon_t^2 = \omega + (\alpha + \beta)\varepsilon_{t-1}^2 + \nu_t - \beta\nu_{t-1}$ (lecture), $\nu_t = \varepsilon_t^2 - \sigma_t^2$ the surprise in the squared shock; ARMA(1,1) with AR coefficient ϕ and MA coefficient θ : $\rho_1 = (1 + \phi\theta)(\phi + \theta)/(1 + 2\phi\theta + \theta^2)$, $\rho_k = \phi^{k-1}\rho_1$; S&P 500 GARCH-N: $\hat{\omega} = 0.0254$, $\hat{\alpha} = 0.1206$, $\hat{\beta} = 0.8599$

▣ Tasks

1. Substitute $\phi = \alpha + \beta$, $\theta = -\beta$ and derive $\rho_1(\varepsilon^2)$.
2. Compute ρ_1 , ρ_{10} and plot the theoretical ACF against the sample ACF of $(r_t - \bar{r})^2$, lags 1–50.
3. Simulate 200 GARCH-N paths of the sample length (500 burn-in days, seed 44) and add the 95% envelope of their sample ACFs.

- ▣ **Report:** the formula, ρ_1 , ρ_{10} , the chart; interpretation: is matching the ACF of r_t^2 a strong or a weak check?

- ▣ **Model:** A1

- ▣ **Notebook:** section A6

B4 [Proposed]: Innovation Laws for Bitcoin — Task

▣ Questions

- ▶ which innovation law describes Bitcoin?
- ▶ is Gaussian-QMLE inference valid there?

- ▣ **Inputs:** Bitcoin closing prices (BTC/USD) from 2014-09-17; returns 2014-09-18 to 2026-09-18, 4 384 returns, 365 days a year; GARCH(1,1), constant mean

▣ Tasks

1. Fit Normal and Student-t innovations; compare ℓ and BIC and draw the QQ plots of $\hat{z}_t$ against each fitted law.
2. Test Normal ($1/\nu = 0$) against Student-t by LR with the boundary law $\frac{1}{2}\chi_0^2 + \frac{1}{2}\chi_1^2$ (5% critical value 2.71).
3. Optional: add skewed-t (ν, λ) and GED (shape ν_G); test $\lambda = 0$ by LR; report every shape parameter with its s.e. and flag active bounds.

- ▣ **Report:** a table of ℓ , BIC, persistence and shape parameters, the QQ chart, the LR tests; interpretation:

- ▶ with $\hat{\nu} < 4$, is Gaussian-QMLE inference valid [Hall & Yao \(2003\)](#)?
- ▶ how does it differ for the Student-t MLE?

- ▣ Model: B1, B2

- ▣ Notebook: section B4

B5 Extended [Solved]: Giacomini–White and the Model Confidence Set — Task

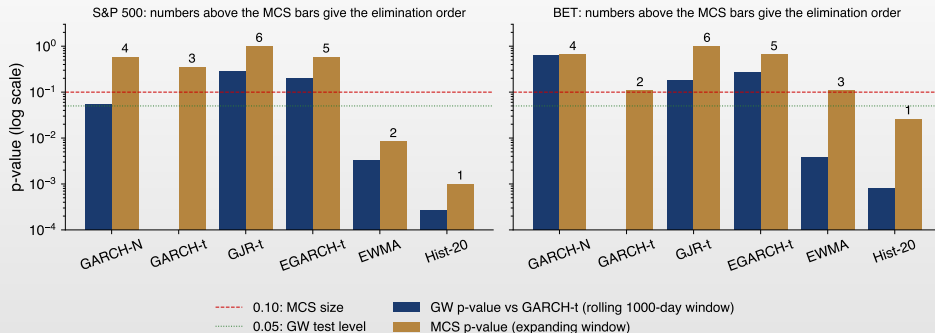
- **Question:** which comparisons of B5 survive once estimated parameters, nested models and six competitors are taken into account?
- **Inputs:** S&P 500 and BET; rolling window of $W = 1\,000$ days refit each January (GW); the expanding-window QLIKE losses of B5 (MCS)
- **Tasks**
 1. For each model m , form $d_{t+1} = L_{t+1}^m - L_{t+1}^{GARCH-t}$ and $Z_{t+1} = (1, d_t)'$; compute $GW = n\bar{Z}'\hat{\Omega}^{-1}\bar{Z}$ with $\hat{\Omega} = n^{-1} \sum Z_{t+1}Z_{t+1}'$ (n : number of forecasts, $\bar{Z}$: mean of Z_{t+1}) and compare with χ_2^2 (5.99) [Giacomini & White \(2006\)](#).
 2. Compute the 90% Model Confidence Set [Hansen, Lunde & Nason \(2011\)](#): statistic T_R (the largest pairwise $|t|$ of loss differences), stationary bootstrap with mean block 10, 10 000 draws, seed 42 (`arch.bootstrap.MCS`).
 3. Report the elimination order and the models kept, for both markets.
- **Report:** GW statistics and p -values, MCS p -values, the chart; interpretation:
 - ▶ why is DM not valid for GJR-t versus GARCH-t [West \(1996\)](#), [Clark & McCracken \(2001\)](#)?
 - ▶ do DM and MCS agree?
- **Notebook:** section B5

B5 Extended [Solved]: Solution

Model	GW S&P	p	MCS p S&P	MCS p BET
GARCH-N	5.85	0.054	0.586	0.658
GARCH-t	—	—	0.353	0.108
GJR-t	2.49	0.288	1.000	1.000
EGARCH-t	3.21	0.201	0.586	0.658
EWMA	11.40	0.003	0.008	0.108
Hist-20	16.41	< 0.001	0.001	0.026

- S&P 500: GW rejects EWMA and Hist-20 only; elimination order Hist-20, EWMA, GARCH-t, GARCH-N, EGARCH-t, GJR-t; kept: the four GARCH models. BET: order Hist-20, GARCH-t, EWMA, GARCH-N, EGARCH-t, GJR-t; the MCS also keeps EWMA ($p = 0.108$)

B5 Extended [Solved]: GW and MCS in a Chart



- Interpretation: under H_0 the nested, expanding-window DM differential degenerates; GW tests the method and stays valid; the MCS controls the error over six models, which pairwise DM tests do not

B6 [Proposed]: Forecast Comparison on the BET — Task

- **Question:** do the B5 conclusions transfer to the Bucharest market?
- **Inputs:** BET, 2 675 forecast days, 2016–2026; the same six models, expanding window, proxy and code as B5
- **Tasks**
 1. Compute the mean QLIKE, the DM t against GARCH- t and the cumulative $\sum d_t$.
 2. Estimate the MZ regression for GARCH- t with HAC errors; report a , b , their s.e., R^2 and the Wald p , and plot the calibration chart.
 3. Put the S&P 500 and BET results side by side, labelling the estimation window of each number.
- **Report:** two tables and two charts; interpretation:
 - ▶ are the BET forecasts calibrated?
 - ▶ what would you test next?
- **Model:** B5
- **Notebook:** section B6

B8 [Proposed]: Bootstrap and IGARCH Test for the BET — Task

▣ Questions

- ▶ how precise is the BET persistence?
- ▶ can its variance have a unit root?

▣ **Inputs:** BET GARCH-t, $\hat{p} = 0.9907$ (half-life 74 days); the two B7 algorithms and functions, $B = 999$

▣ Tasks

1. Run Algorithm 1 and report the percentile interval for p , the half-life interval and the share of draws at $p = 1$.
2. Compare the bootstrap s.d. of $\hat{p}^*$ with the robust s.e. of $\hat{\alpha} + \hat{\beta}$.
3. Run Algorithm 2 and compare the bootstrap, mixture and χ^2_1 p -values at 5%.

▣ **Report:** the intervals, the two s.e., the test and the two-panel chart; interpretation: do the separate BET and S&P 500 intervals rank the two half-lives?

▣ **Model:** B7

▣ **Notebook:** section B8

C1 [Proposed]: Does US News Drive BET Volatility? — Task (1/2): Data and Timing

▣ Questions

- ▶ does yesterday's S&P 500 shock raise today's BET variance?
- ▶ by how much?

▣ Inputs: BET and S&P 500 closing prices, 2000-01-07 to 2026-09-18; align the *prices* on common days, then compute returns; t indexes the common days

▣ Tasks

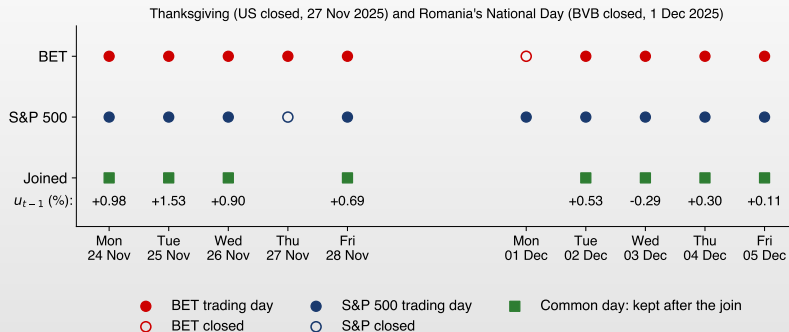
1. Build the sample of common days and u_{t-1} , the S&P 500 return of the previous common day (Wall Street closes after Bucharest).
2. Check the holiday week below: on 2 December 2025, u_{t-1} is the S&P 500 return from 26 to 28 November (0.69%); say which US session is missing and where it goes.

▣ Report: the number of common days and the timing check

▣ Model: B1, B2

▣ Notebook: section C1

C1 [Proposed]: the Common-Day Calendar around a Holiday



□ Filled markers: trading days; squares: days kept after aligning the prices; the numbers are u_{t-1} for each common day

C1 [Proposed]: Task (2/2): Model, Test, Contribution

- ▣ **Questions:** the same questions, now as an estimated model
- ▣ **Inputs:** $r_t = \mu + \phi u_{t-1} + \varepsilon_t$, $\sigma_t^2 = \omega + \alpha \varepsilon_{t-1}^2 + \beta \sigma_{t-1}^2 + \delta u_{t-1}^2$, $\varepsilon_t = \sigma_t z_t$ with standardised Student-t z_t ; notebook functions `spill_negll`, `fit_spill`
- ▣ **Tasks**
 1. Write the log-likelihood step by step (start $\sigma_1^2 = \text{sample variance}$) and estimate the unrestricted model and the model with $\delta = 0$.
 2. Test $\delta = 0$ by LR; since $\delta \geq 0$, use $\frac{1}{2}\chi_0^2 + \frac{1}{2}\chi_1^2$; repeat on 2000–2012 and 2013–2026.
 3. Compute the accumulated US term $s_t = \beta s_{t-1} + \delta u_{t-1}^2$ with $s_1 = 0$, the share s_t/σ_t^2 and the volatility without it, $\sqrt{\sigma_t^2 - s_t}$; plot both.
- ▣ **Report:** estimates for three samples, the LR tests, the contribution chart; a Quantlet folder and one slide with the conclusion and its main limitation; interpretation: is the link causal or predictive?

C3 [Proposed]: Quasi Score-Driven GARCH on US Stocks — Task (1/2)

- **Question:** is the tail thickness of returns the same thing as the downweighting of large shocks in the variance update Blasques, Francq & Laurent (2023)?
- **Inputs:** start with JPMorgan (adjusted closing prices) and the VIX (closing values), aligned on common days, 3 January 1995 – 28 February 2019; $r_t = 100 \Delta \ln P_t$, $X_t = \text{VIX}_t^2/252$ (the VIX of day t enters f_{t+1})
- **Tasks**
 1. Model (the estimated recursion): $y_t = r_t - \mu - \phi r_{t-1} = \sqrt{f_t} \epsilon_t$, ϵ_t standardised Student-t with $1/\xi$ degrees of freedom; $f_{t+1} = \omega + \varpi X_t + \alpha \Psi\left(\frac{1+\zeta}{1-2\zeta+\zeta\epsilon_t^2}\right) \epsilon_t^2 f_t + \beta f_t$, $\Psi(x) = x \tanh(cx/2)$, $c = 1000$, $\zeta \in (-1, 1/2)$, f_1 = variance of y_t ; ϖ : weight of the VIX term X_t .
 2. Parameters: ξ = tail thickness, ζ = downweighting; $\zeta = 0$: GARCH(1,1)-t; $\zeta = \xi$: Beta-t-GARCH. Estimate the three models by maximum likelihood (notebook of MFM_ch5_quasi_score_driven: qsd_negll, fit_qsd).
 3. Test $\zeta = 0$ and $\xi = \zeta$ by LR against χ_1^2 ($\zeta = 0$ is interior thanks to Ψ).
- **Report:** estimates, log-likelihoods, the two LR tests
- **Model:** B1, B2

C3 [Proposed]: Task (2/2): Diagnostics and Extension

▣ Questions

- ▶ do the three models fit?
- ▶ does the answer hold beyond one stock?

- ▣ **Inputs:** the fits of Task 1; PIT $u_t = F(\epsilon_t)$, F the fitted distribution function of ϵ_t , i.i.d. $U(0, 1)$ if the model is right [Diebold, Gunther & Tay \(1998\)](#); T : sample length

▣ Tasks

1. Plot $\sqrt{f_t}$ of the three models around the largest shock and the PIT histograms.
2. Compute the naive statistic $(\sum_t u_t - T/2)/\sqrt{T/12}$ and explain why it ignores estimation error (the correction of [Pierce \(1982\)](#) lowers the variance below $T/12$).
3. Optional: extend to the 11 course stocks and to 18 September 2026; report rejection frequencies with a binomial interval that assumes, hypothetically, independent rejections.

- ▣ **Report:** a Quantlet folder and one slide with the table, the tests and the main limitation; interpretation: why can 11 selected survivors not confirm the paper's 89.5%?

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